

EEE-301

Data and Telecommunication

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Chapter-4

Digital Transmission

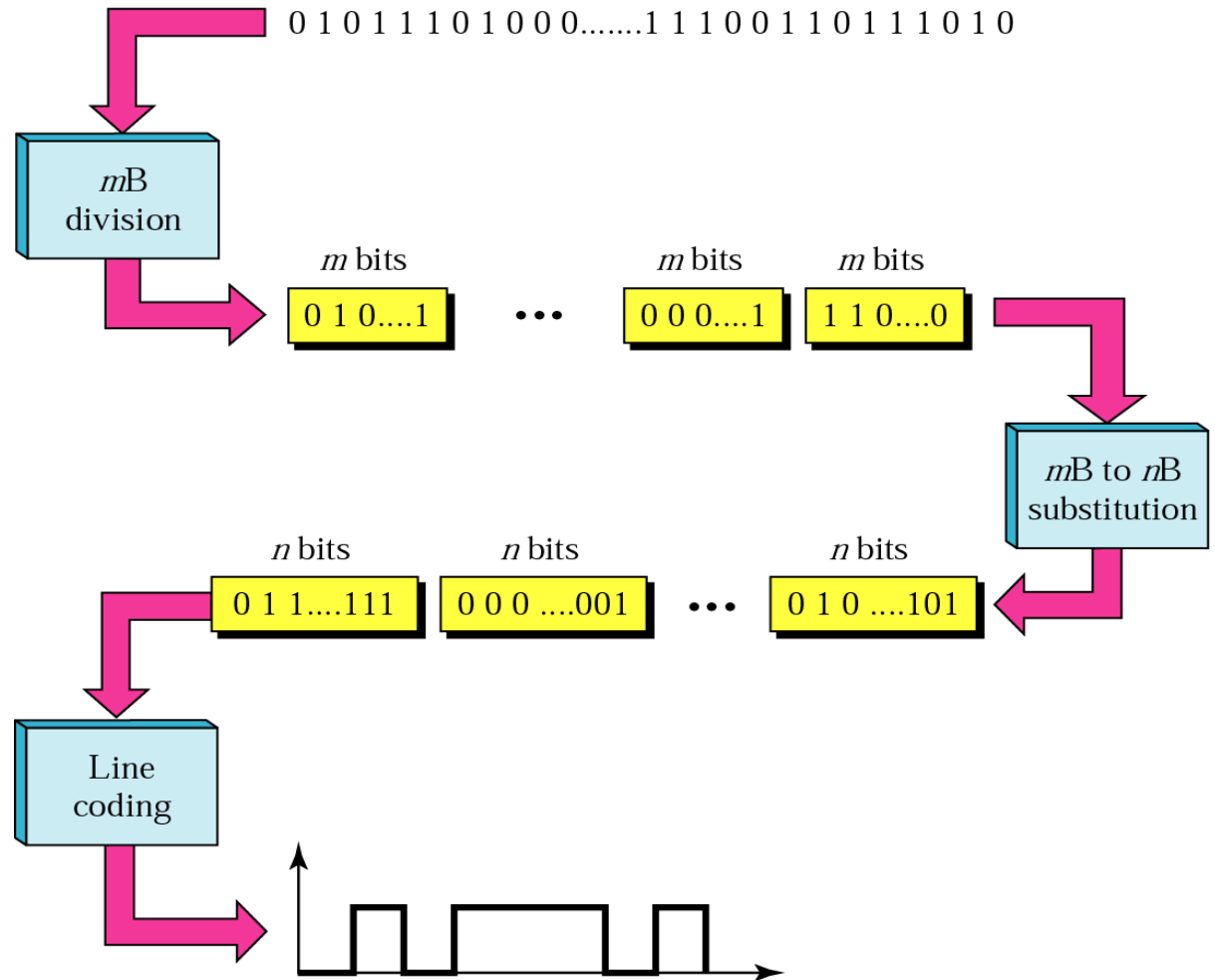
Textbook: Data Communications and Networking
Fourth Edition
- Behrouz A. Forouzan

Digital Transmission

- **Data transmission** also **data communication** or **digital communications** is the transfer of data (a digital bitstream or a digitized analog signal) over a point-to-point or point-to-multipoint communication channel.
- Examples of such channels are copper wires, optical fibers, wireless communication channels, storage media and computer buses.
- The data are represented as an electromagnetic signal, such as an electrical voltage, radiowave, microwave, or infrared **signal**.

Bit to Digital Signal Coding

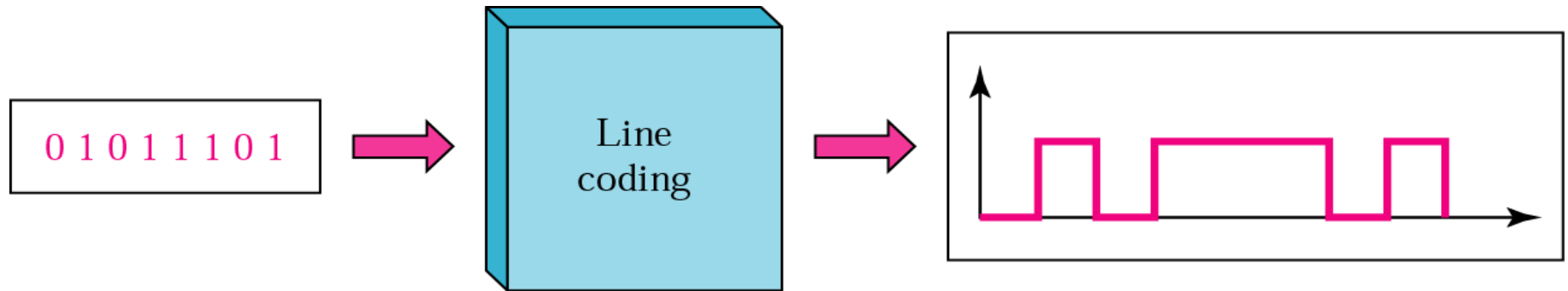
- Line Coding
- Block Coding



Line Coding

- The process for converting digital data into digital signal is said to be Line Coding.
- Digital data is found in binary format.
- It is represented (stored) internally as series of 1s and 0s.

Line Coding



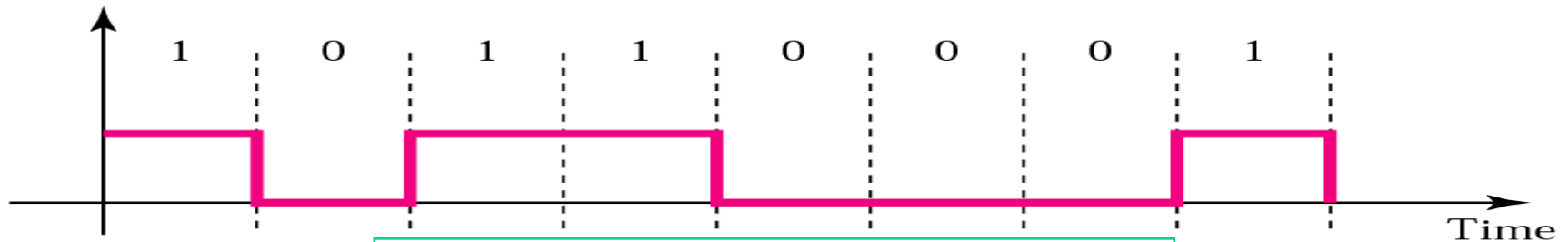
Some **characteristics** of line coding

- signal level vs. data level
- pulse rate vs. bit rate
- dc components
- self-synchronization

Signal Level vs. Data Level

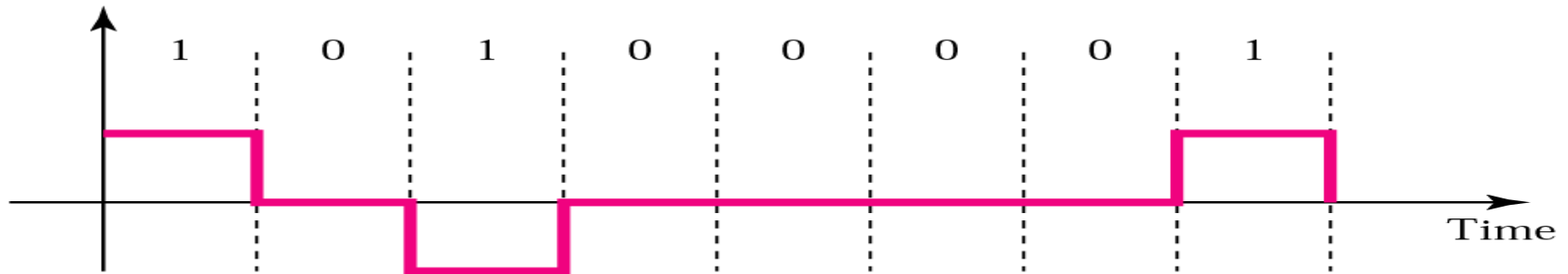
- The number of different values allowed in a signal is the signal level.
- The number of symbols that represent data is the data level.

Amplitude



a. Two signal levels and two data levels

Amplitude



b. Three signal levels and two data levels

Pulse Rate vs. Bit Rate

- Pulse rate = number of pulses per second
 - A pulse = minimum time to transmit a symbol
- If a pulse contains 1 bit $\Rightarrow$ pulse rate equals bit rate
- In general
 - $\text{BitRate} = \text{PulseRate} \times \log_2 L$
 - L: number of data levels

Pulse Rate vs. Bit Rate

Example

A signal has two data levels with a pulse duration of 1 ms. We calculate the pulse rate and bit rate as follows:

$$\text{Pulse Rate} = 1 / 10^{-3} = 1000 \text{ pulses/s}$$

$$\text{Bit Rate} = \text{Pulse Rate} \times \log_2 L = 1000 \times \log_2 2 = 1000 \text{ bps}$$

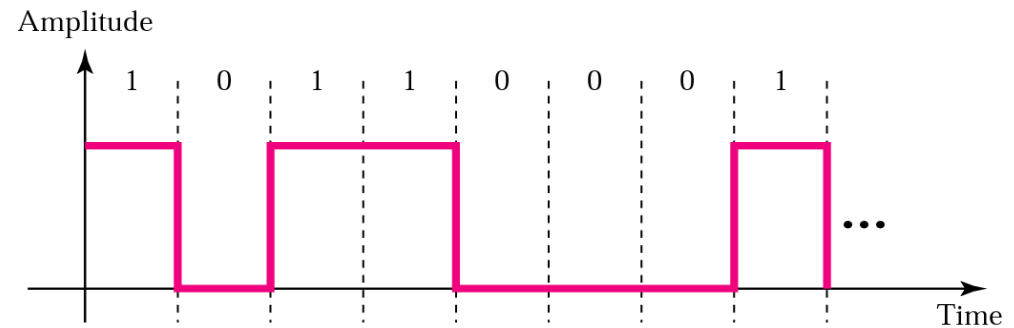
Question: A signal has four data levels with a pulse duration of 1 ms. What are the pulse and bit rates?

DC (direct-current) Component

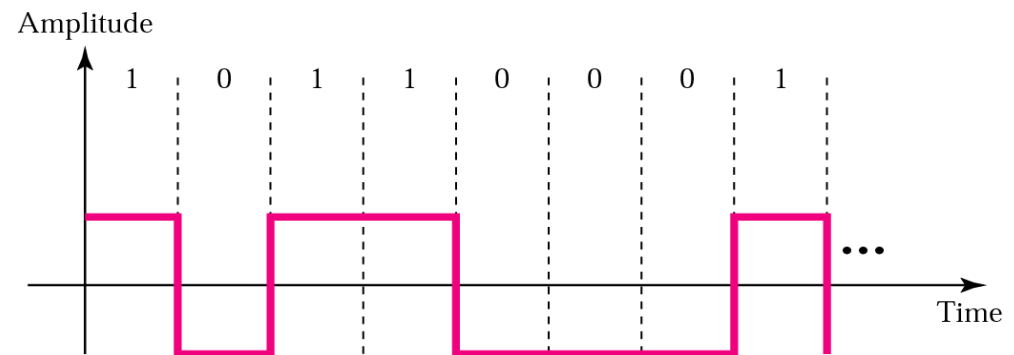
DC component if avg.
amplitude $>$ zero

If signal with a dc
component is to pass through
a transformer that does not
pass a dc component $\Rightarrow$ error
in output signal

dc component is extra energy
on the line but useless



a. A signal with dc component

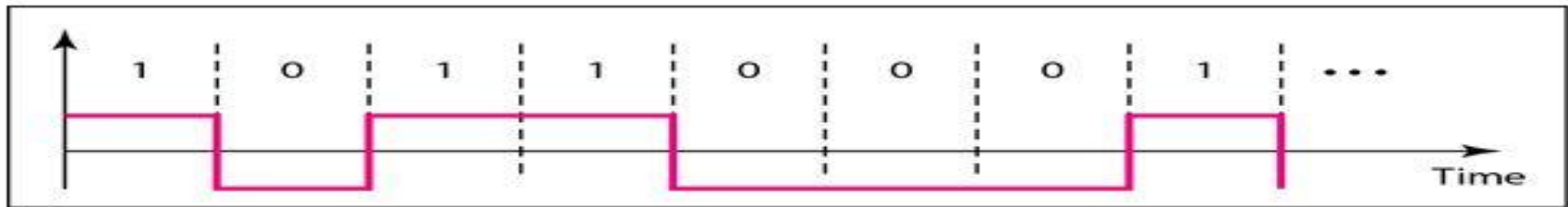


b. A signal without dc component

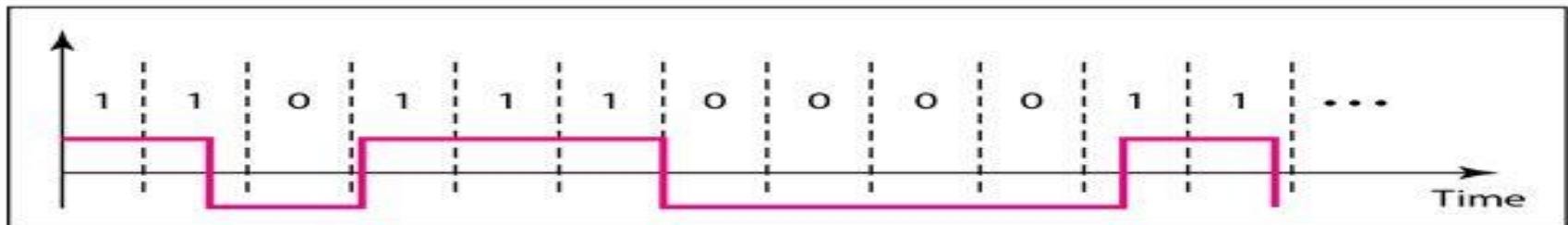
A digital signal should not have a dc component

Self-synchronization

- To correctly interpret the signals received from the sender, the receiver's bit intervals must correspond exactly to the sender's bit intervals.
- If the receiver clock is faster or slower, the bit intervals are not matched and the receiver might misinterpret the signals.
- The following figure represents the synchronization problem.



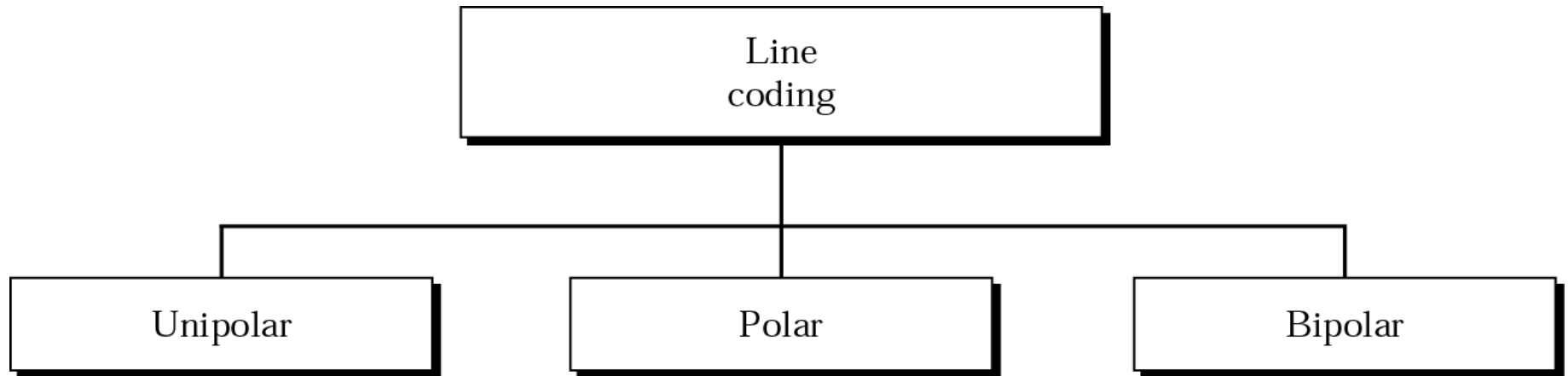
a. Sent



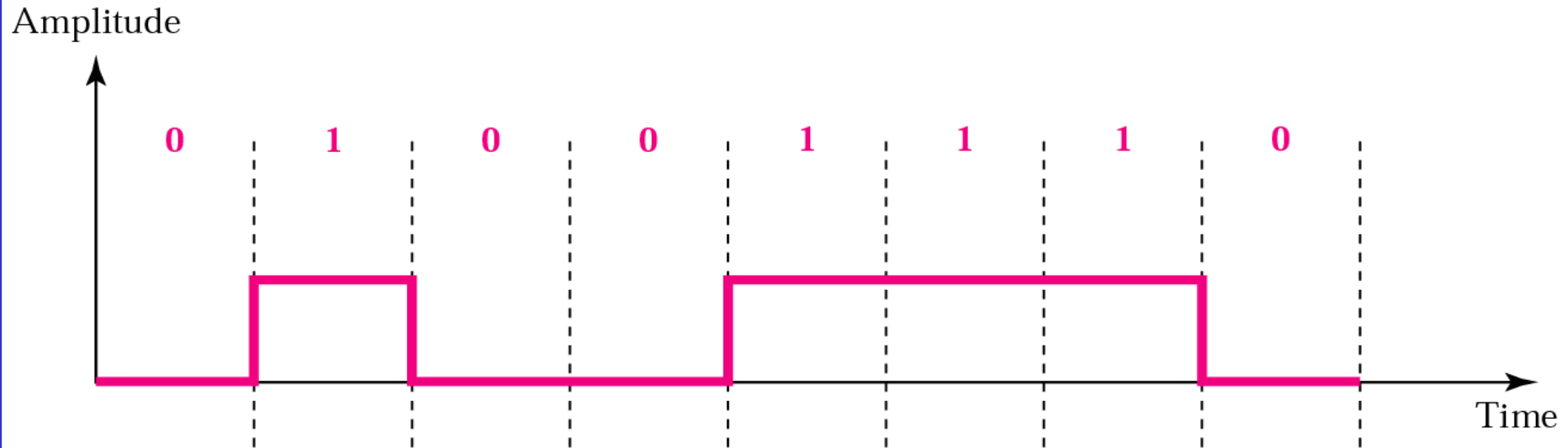
b. Received

Line Coding Schemes

Digital signal is denoted by discrete signal, which represents digital data. There are three types of line coding schemes available:



Unipolar Encoding

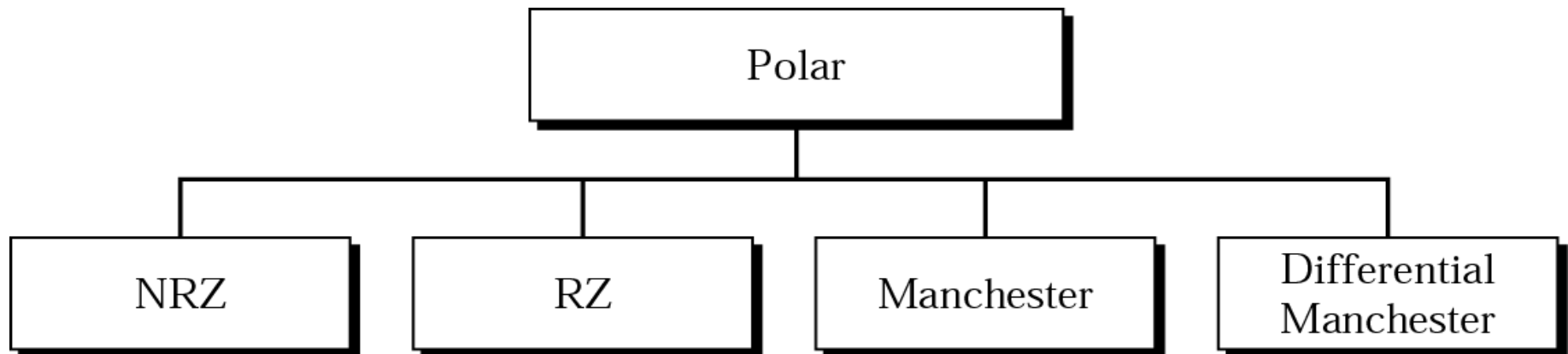


Using only one voltage level

Unipolar encoding schemes use single voltage level to represent data. In this case, to represent binary 1, high voltage is transmitted and to represent 0, no voltage is transmitted. It is also called Unipolar-Non-return-to-zero, because there is no rest condition i.e. it either represents 1 or 0.

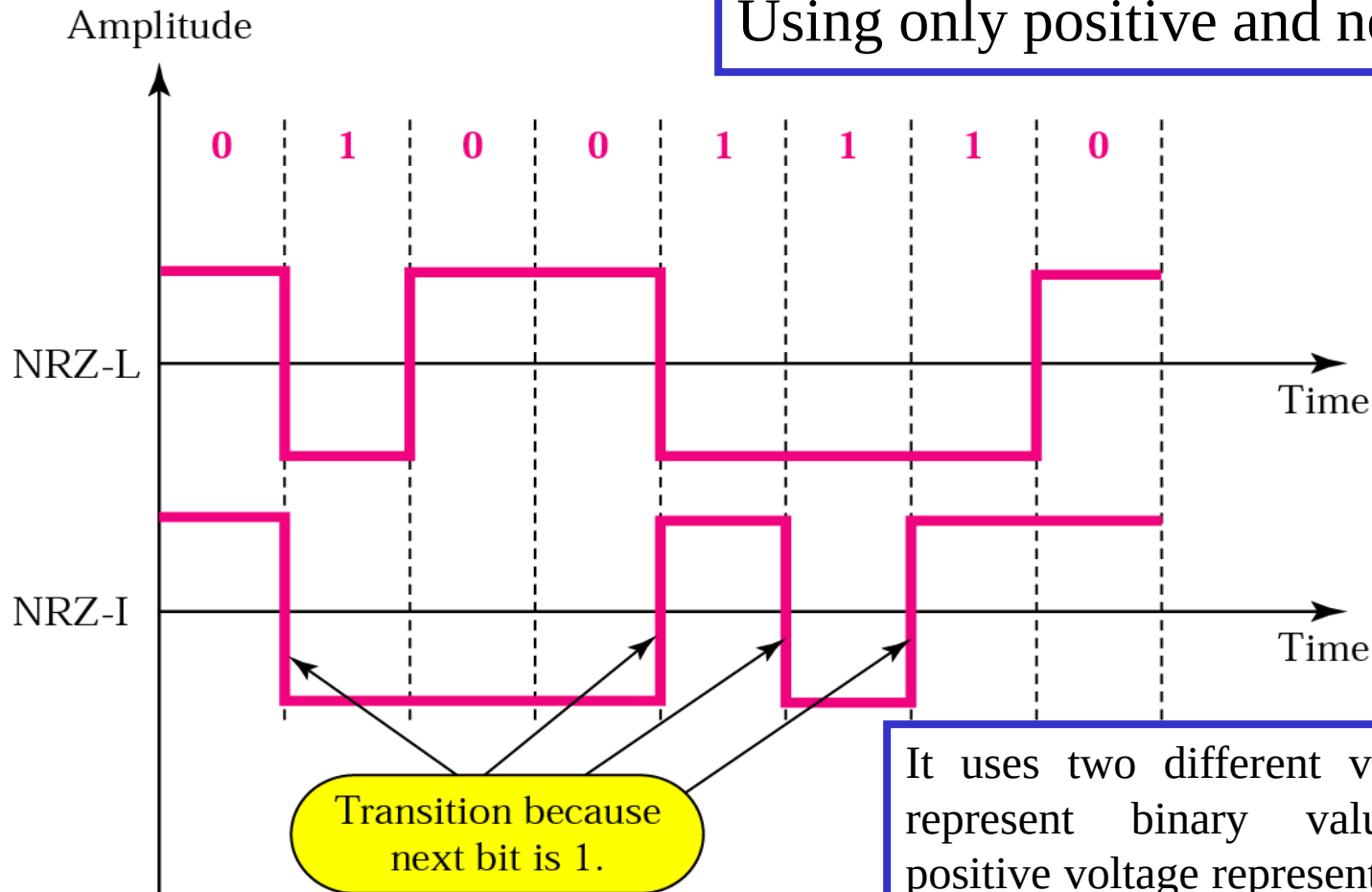
Polar Encoding

*Using two voltage levels:
positive and negative*



NRZ: Non-Return to Zero

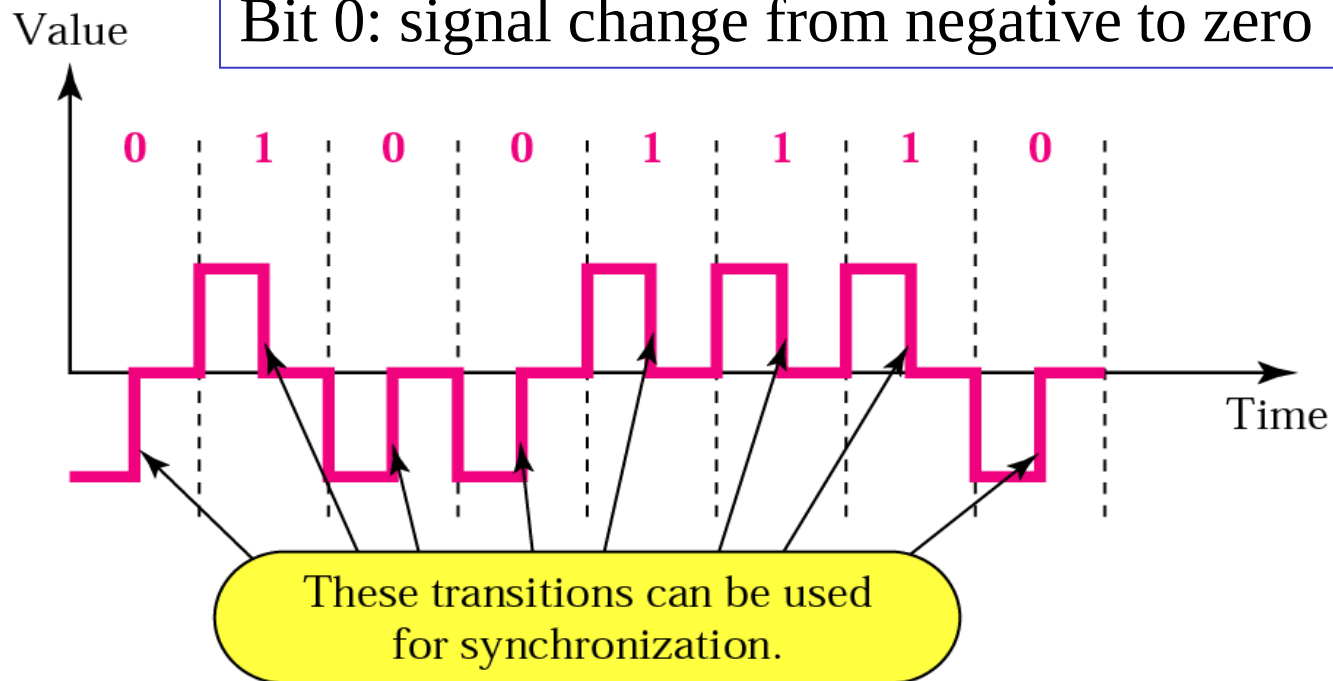
Using only positive and negative values



RZ: Return to Zero

Bit 1: signal change from positive to zero

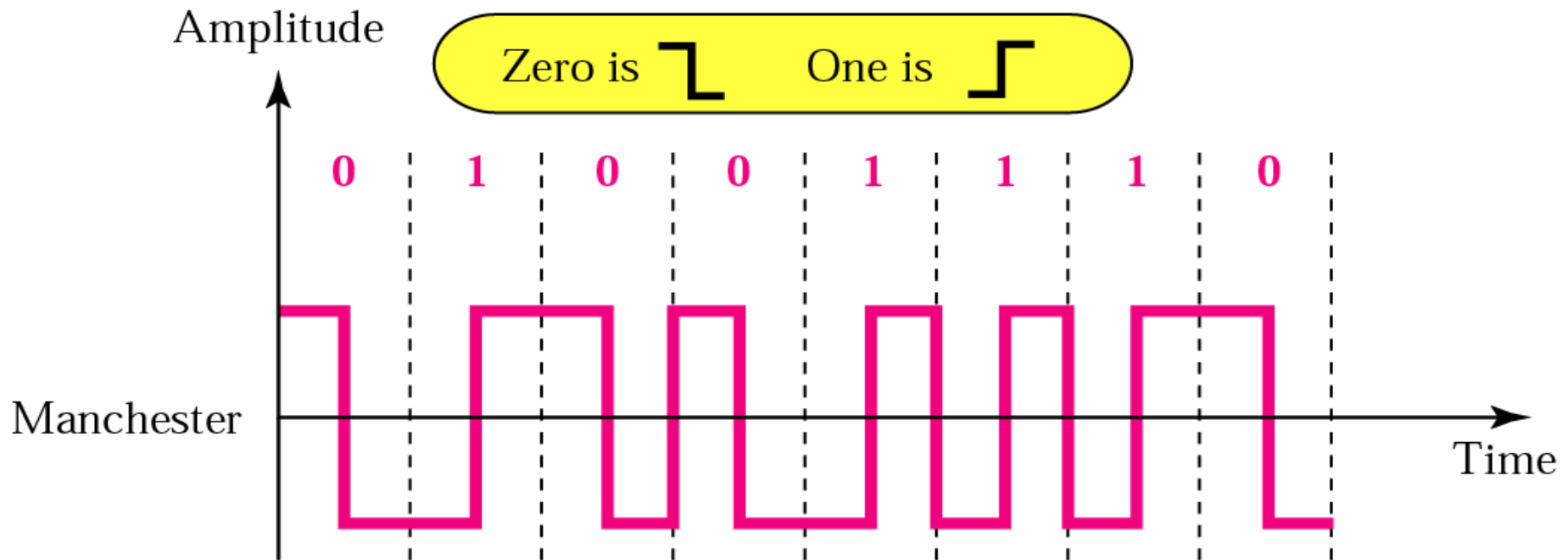
Bit 0: signal change from negative to zero



Advantage: ensure synchronization

Disadvantage: require 3 levels of amplitude => more bandwidth

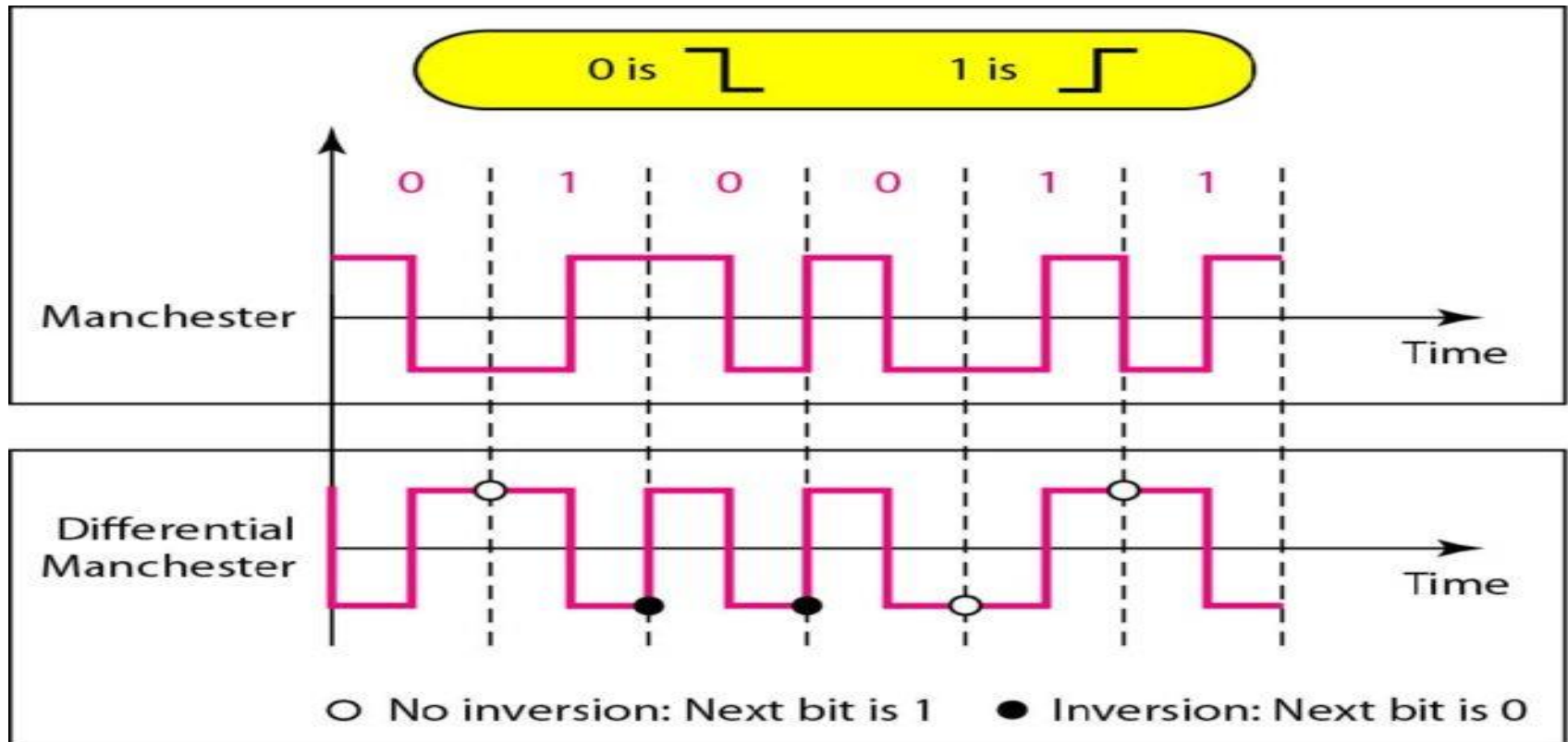
Manchester



Self-synchronizable
Use only 2 levels of amplitudes

Differential Manchester

- In this type of coding, there always occurs a transition in the middle of the bit interval. If there occurs a transition at the beginning of the bit interval, then the input bit is 0. If no transition occurs at the beginning of the bit interval, then the input bit is 1.



Bi-Polar Encoding

- In bipolar encoding (sometimes called multilevel binary), there are three voltage levels: positive, negative, and zero.
- The voltage level for one data element is at zero.
- While the voltage level for the other element alternates between positive and negative.

Bipolar Encoding

- Two variations of bipolar encoding: AMI and pseudoternary.
- A common bipolar encoding scheme is called bipolar alternate mark inversion (AMI).
- In the term alternate mark inversion, the word mark comes from telegraphy and means 1.
- So AMI means alternate 1 inversion.
- A neutral zero voltage represents binary 0.
- Binary 1s are represented by alternating positive and negative voltages.

Bipolar Encoding

- AMI is commonly used for long-distance communication, but it has a synchronization problem when a long sequence of 0s is present in the data.
- A variation of AMI encoding is called pseudoternary in which the 1 bit is encoded as a zero voltage and the 0 bit is encoded as alternating positive and negative voltages.

Bipolar Encoding

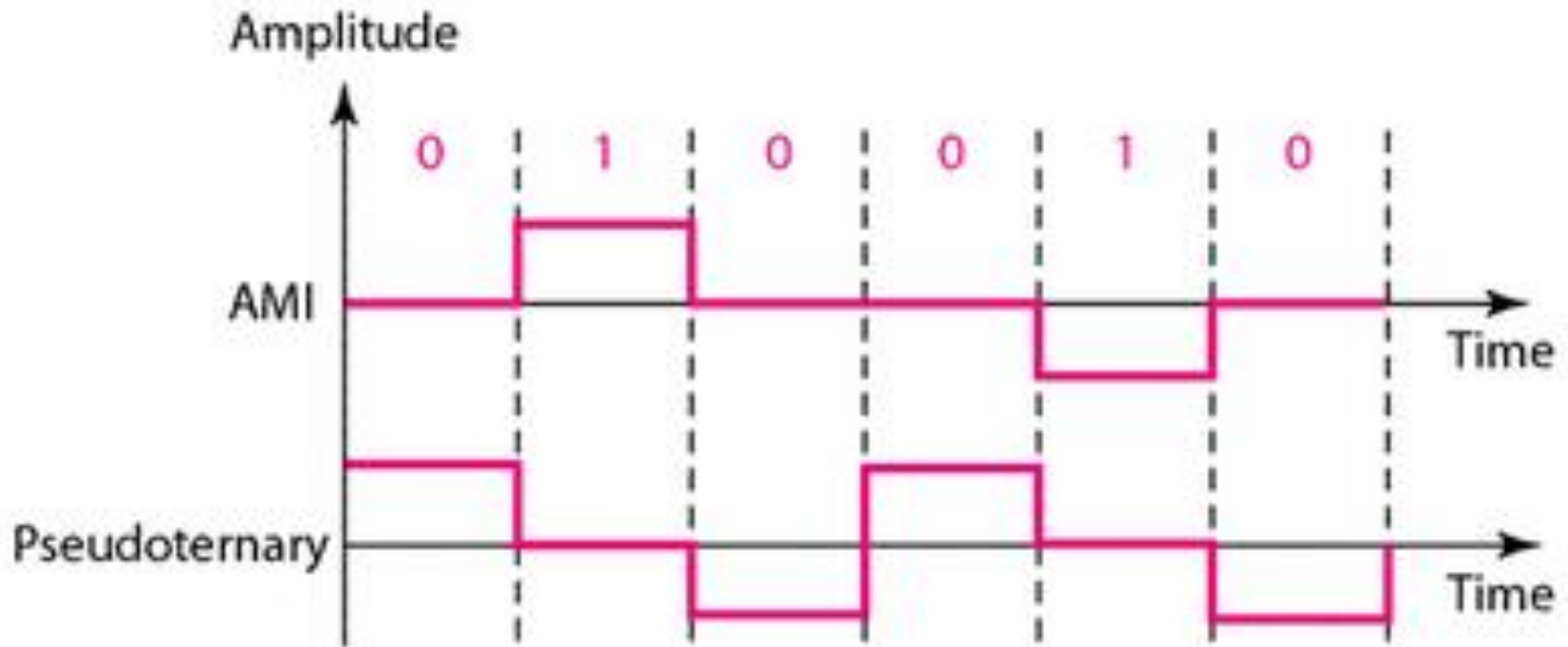


Figure: Bipolar Encoding (AMI and Pseudoternary)

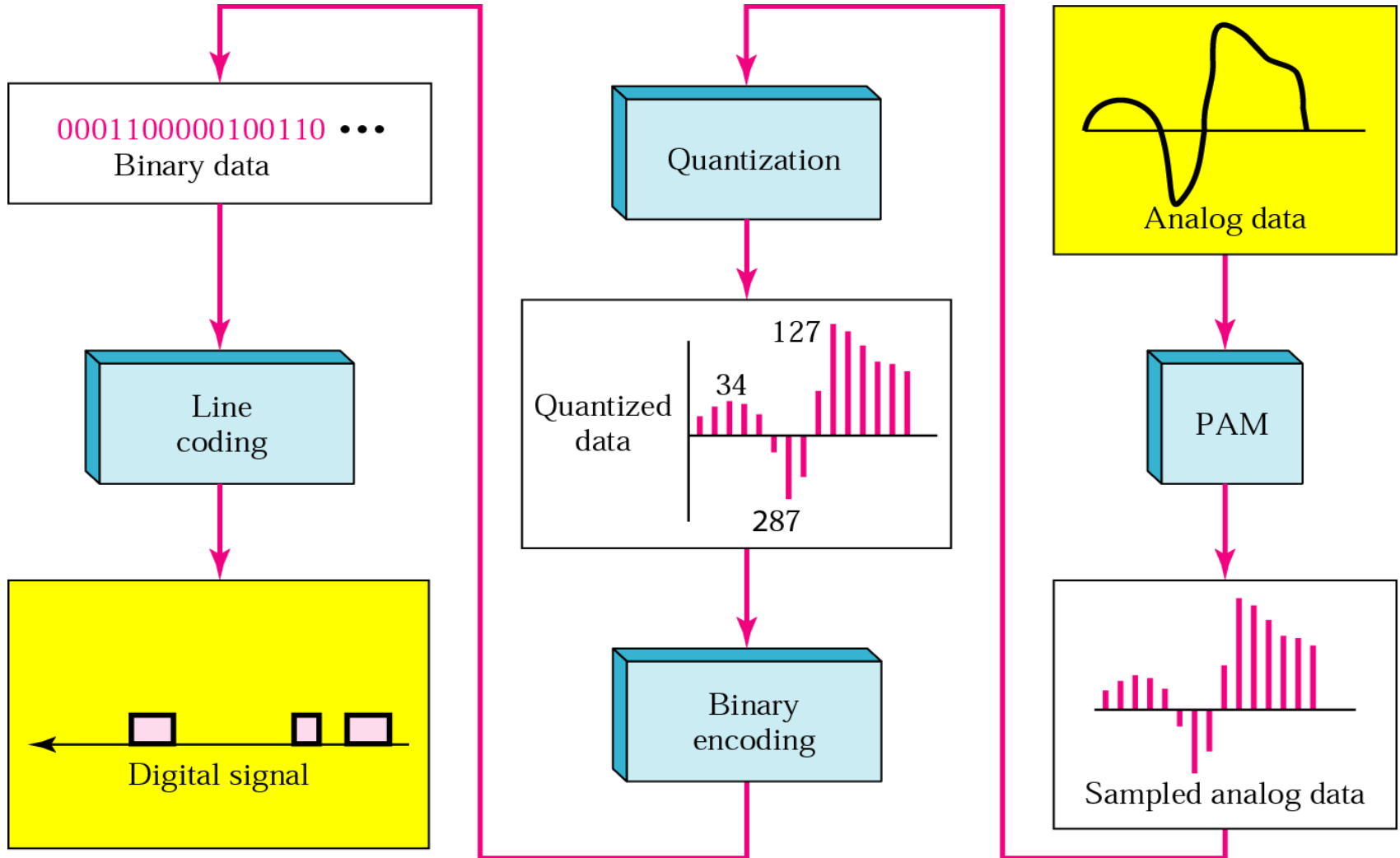
ANALOG-TO-DIGITAL CONVERSION

- Two techniques- pulse code modulation and delta modulation.

Pulse Code Modulation (PCM):

- The most common technique to change an analog signal to digital data (digitization) is called pulse code modulation (PCM).
- A PCM encoder has three processes
 - The analog signal is sampled.
 - The sampled signal is quantized.
 - The quantized values are encoded as streams of bits.

ANALOG-TO-DIGITAL CONVERSION

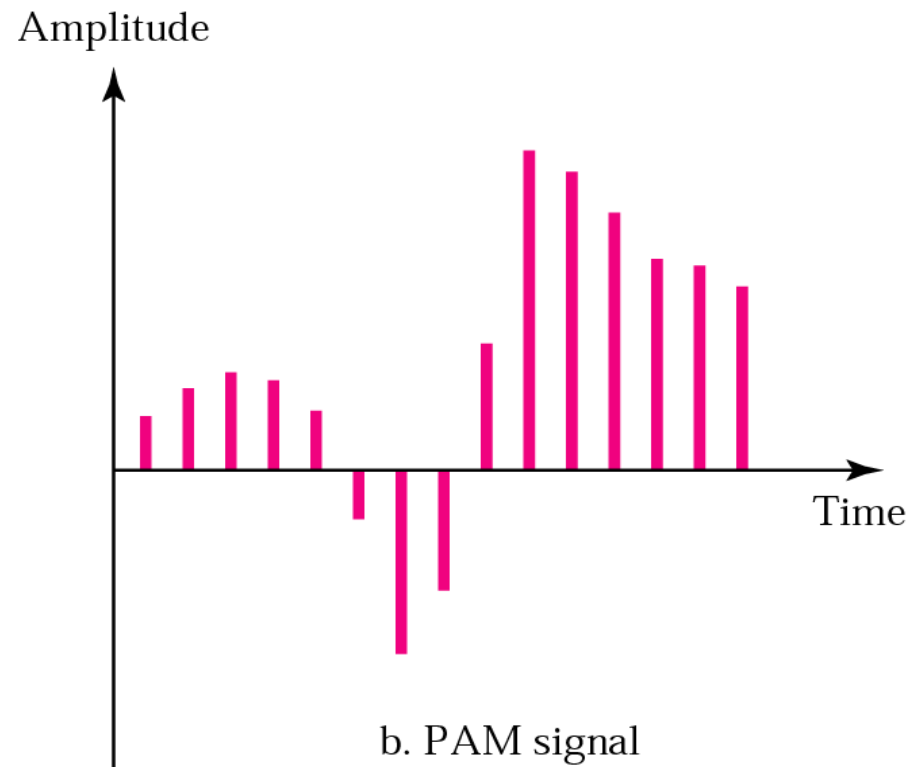
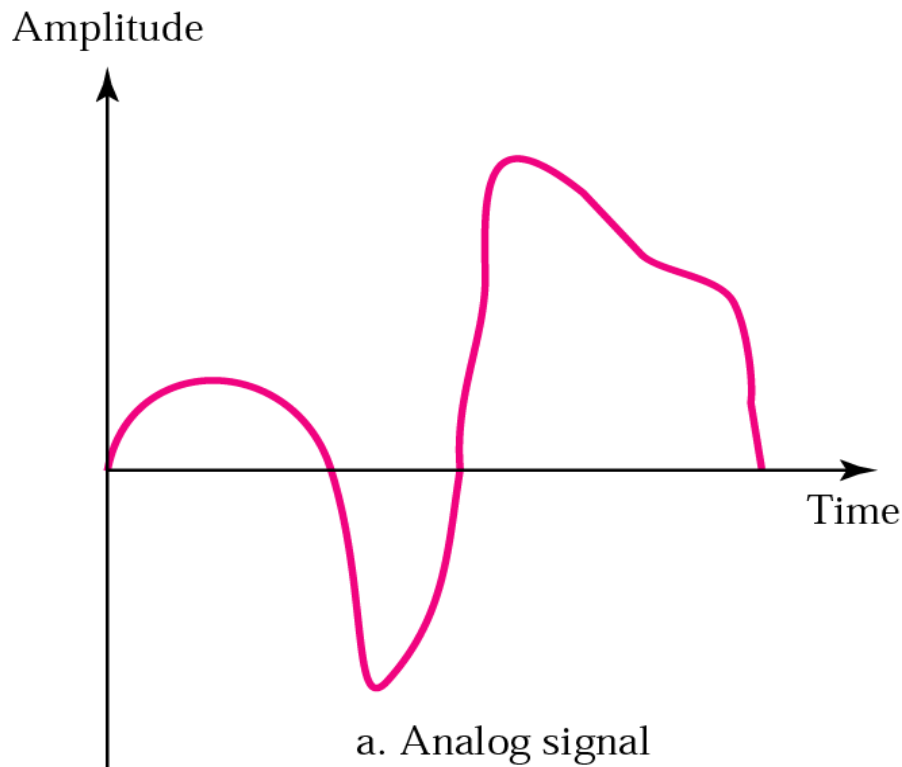


Sampling: Analog to digital

- The first step in PCM is sampling.
- The analog signal is sampled every T_s , where T is the sample interval or period.
- The inverse of the sampling interval is called the sampling rate or sampling frequency.

Pulse Amplitude Modulation (PAM)

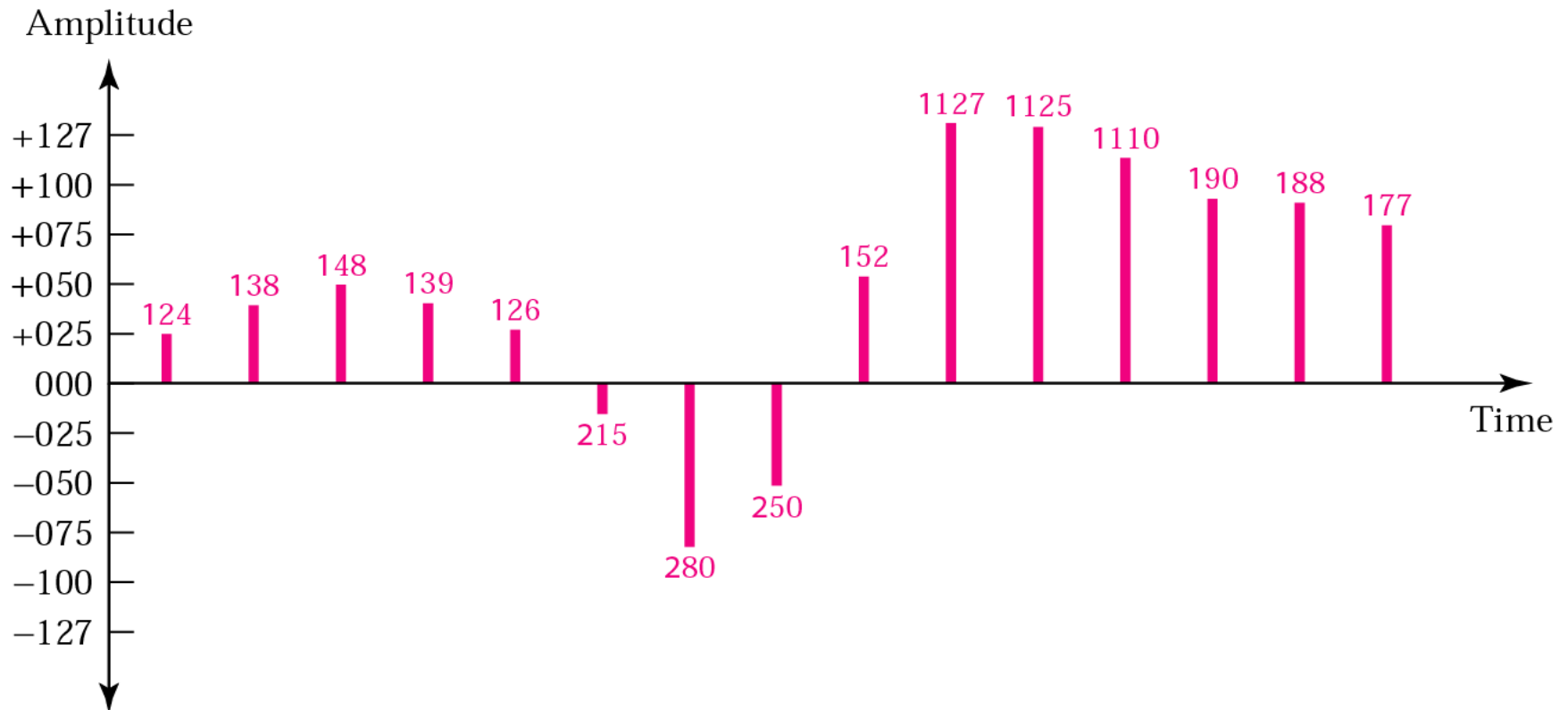
One analog-to-digital conversion method. PAM has some applications, but it is not used by itself in data communication. However, it is the first step in another very popular conversion method called pulse code modulation.



Quantized PAM

- Quantization

Method of assigning integral values in a specific range to sampled instance.



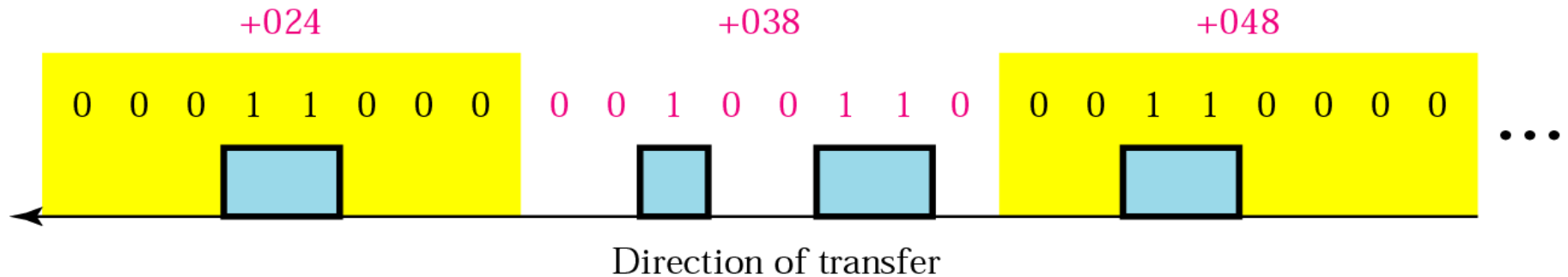
Quantizing by using sign and magnitude

+024	00011000	-015	10001111	+125	01111101
+038	00100110	-080	11010000	+110	01101110
+048	00110000	-050	10110010	+090	01011010
+039	00100111	+052	00110110	+088	01011000
+026	00011010	+127	01111111	+077	01001101

Sign bit
+ is 0 - is 1

Pulse Code Modulation

PCM modifies the pulses created by PAM to create a completely digital signal.



Sampling Rate: Nyquist Theorem

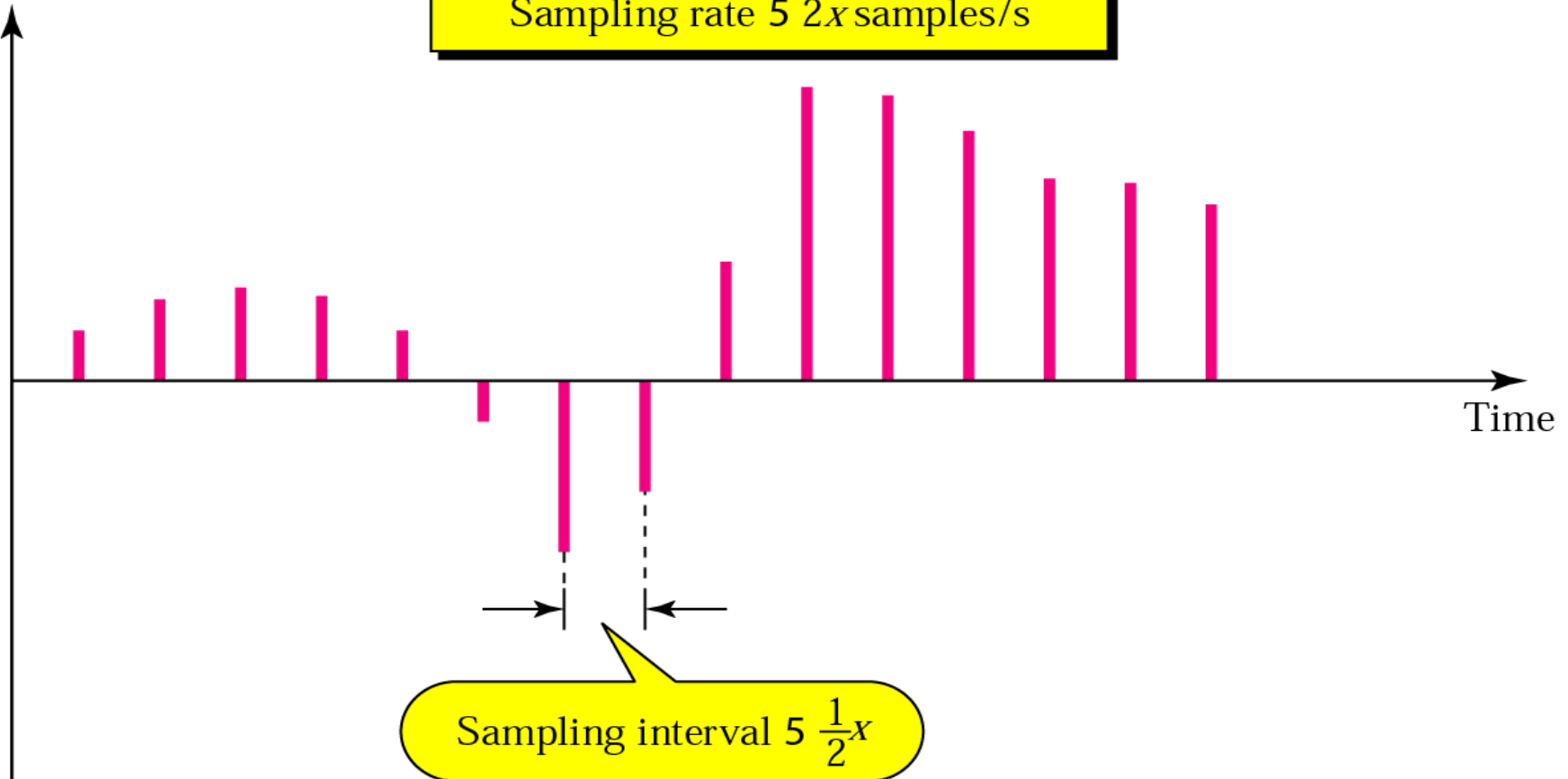
sampling rate must be at least 2 times the highest frequency

Amplitude

Highest frequency $5 \times \text{Hz}$
Sampling rate $5 \times 2 \times \text{samples/s}$

Time

Sampling interval $5 \times \frac{1}{2} \times$



Question

What sampling rate is needed for a signal with a bandwidth of 10,000 Hz (1000 to 11,000 Hz)?

The sampling rate must be twice the highest frequency in the signal:

$$\text{Sampling rate} = 2 \times (11,000) = 22,000 \text{ samples/s}$$

More Question

We want to digitize the human voice. What is the bit rate, assuming 8 bits per sample? The human voice normally contains frequencies from 0 to 4000 Hz.

$$\text{Sampling rate} = 4000 \times 2 = 8000 \text{ samples/s}$$

$$\begin{aligned} \text{Bit rate} &= \text{sampling rate} \times \text{number of bits per sample} \\ &= 8000 \times 8 = 64,000 \text{ bps} = 64 \text{ Kbps} \end{aligned}$$

Block Coding

To ensure accuracy of the received data frame redundant bits are used. For example, in even-parity, one parity bit is added to make the count of 1s in the frame even. This way the original number of bits is increased. It is called Block Coding.

Block coding is represented by slash notation, mB/nB . Means, m -bit block is substituted with n -bit block where $n > m$. Block coding involves three steps:

Division,

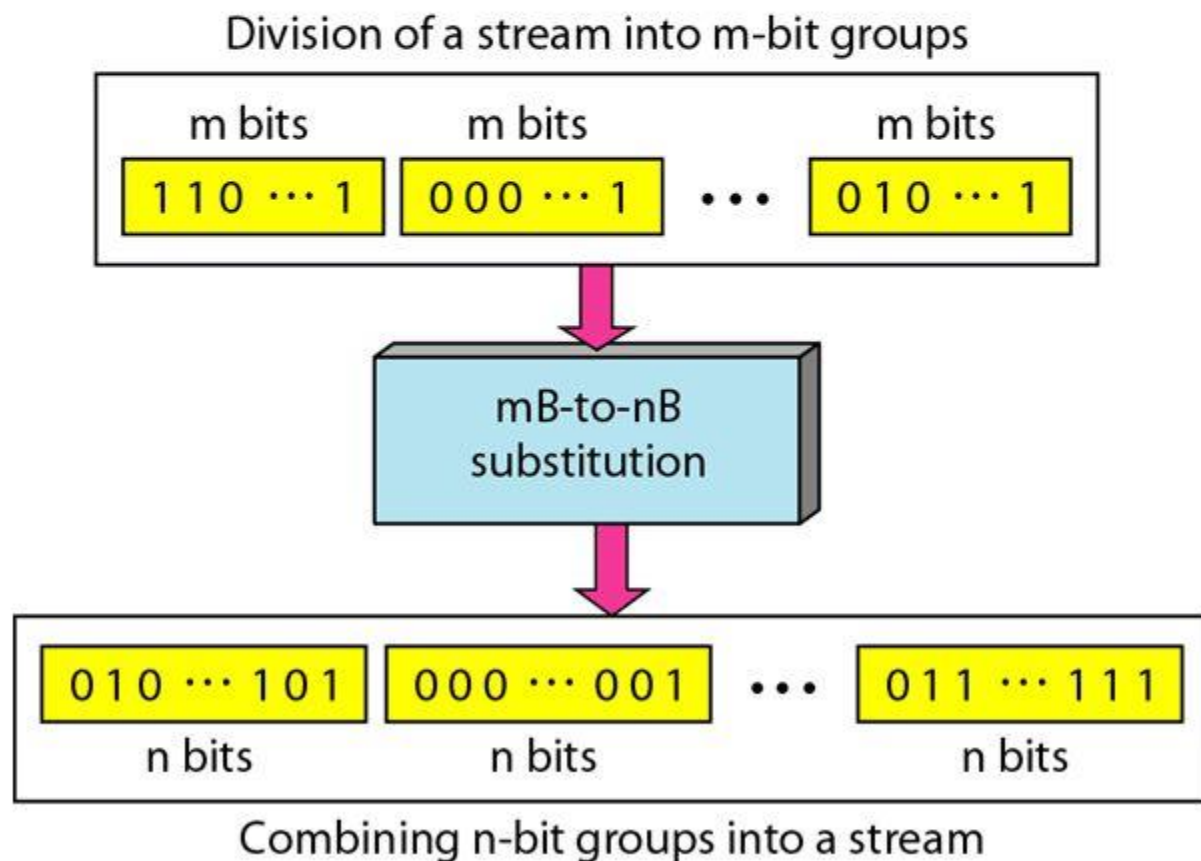
Substitution

Combination.

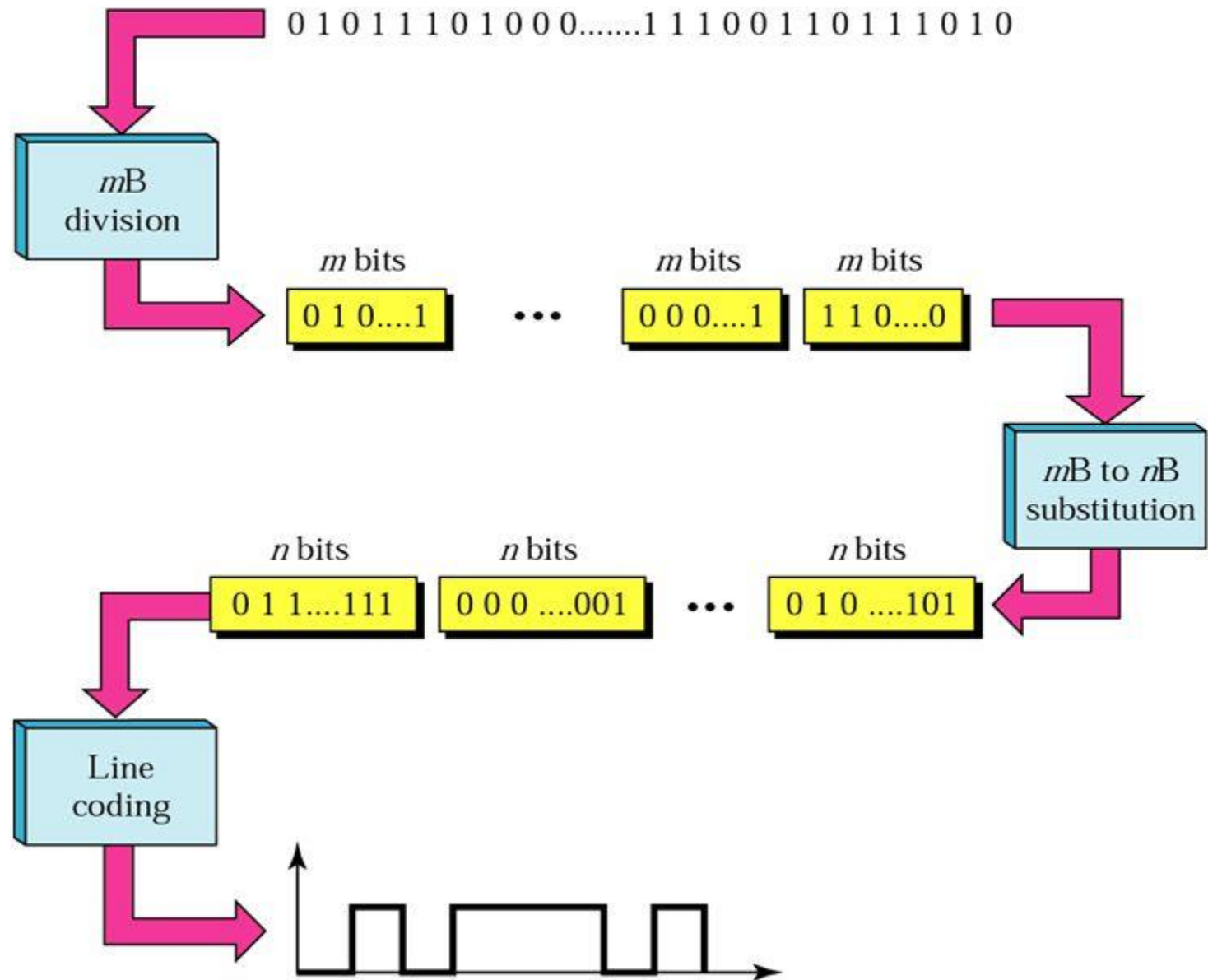
After block coding is done, it is **line coded** for transmission.

Block coding concept

Block coding is normally referred to as mB/nB coding; it replaces each m-bit group with an n-bit group.



Block coding

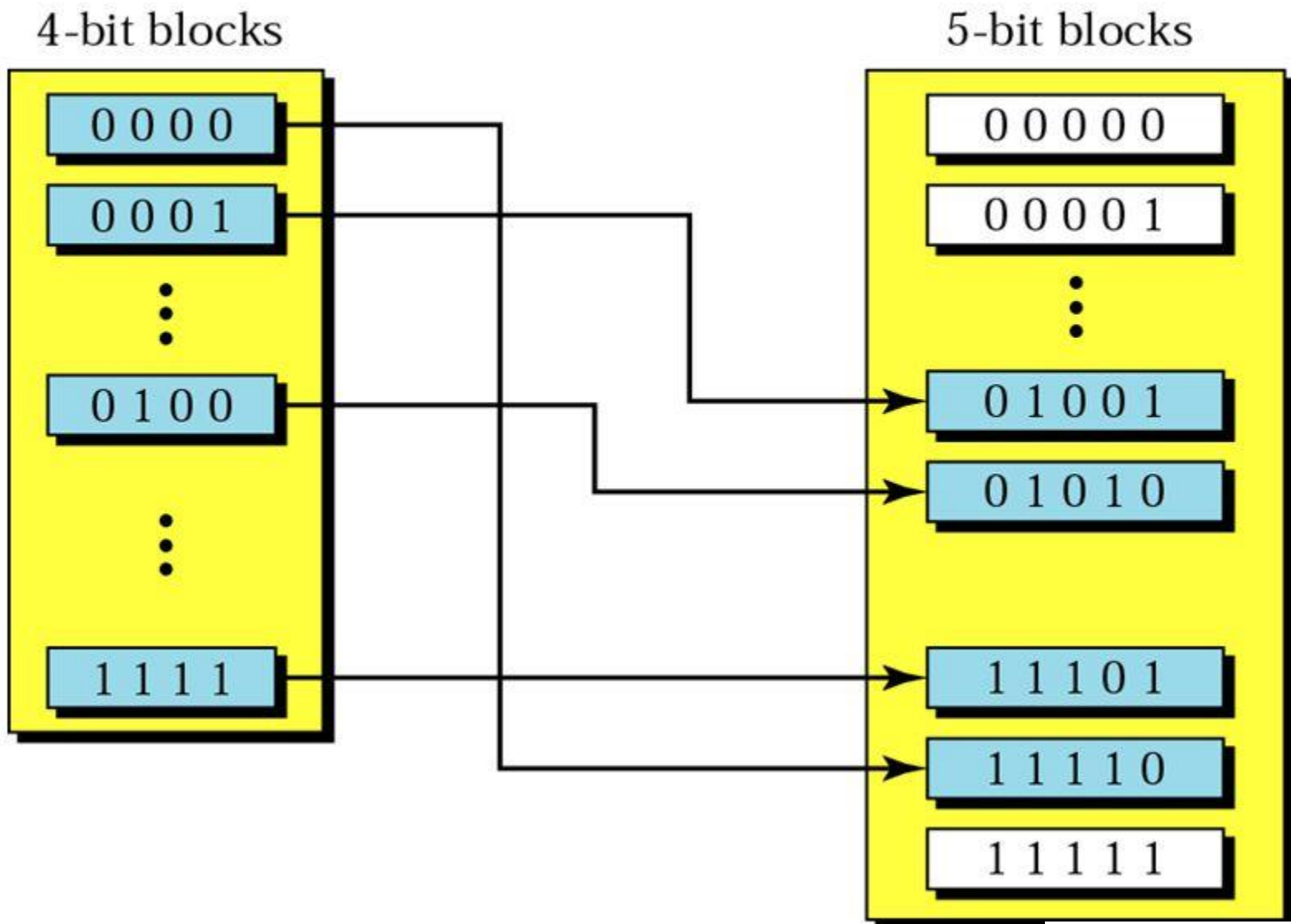


Block Coding

<i>Data Sequence</i>	<i>Encoded Sequence</i>	<i>Control Sequence</i>	<i>Encoded Sequence</i>
0000	11110	Q (Quiet)	00000
0001	01001	I (Idle)	11111
0010	10100	H (Halt)	00100
0011	10101	J (Start delimiter)	11000
0100	01010	K (Start delimiter)	10001
0101	01011	T (End delimiter)	01101
0110	01110	S (Set)	11001
0111	01111	R (Reset)	00111
1000	10010		
1001	10011		
1010	10110		
1011	10111		
1100	11010		
1101	11011		
1110	11100		
1111	11101		

Figure: 4B/5B encoding

Substitution in block coding

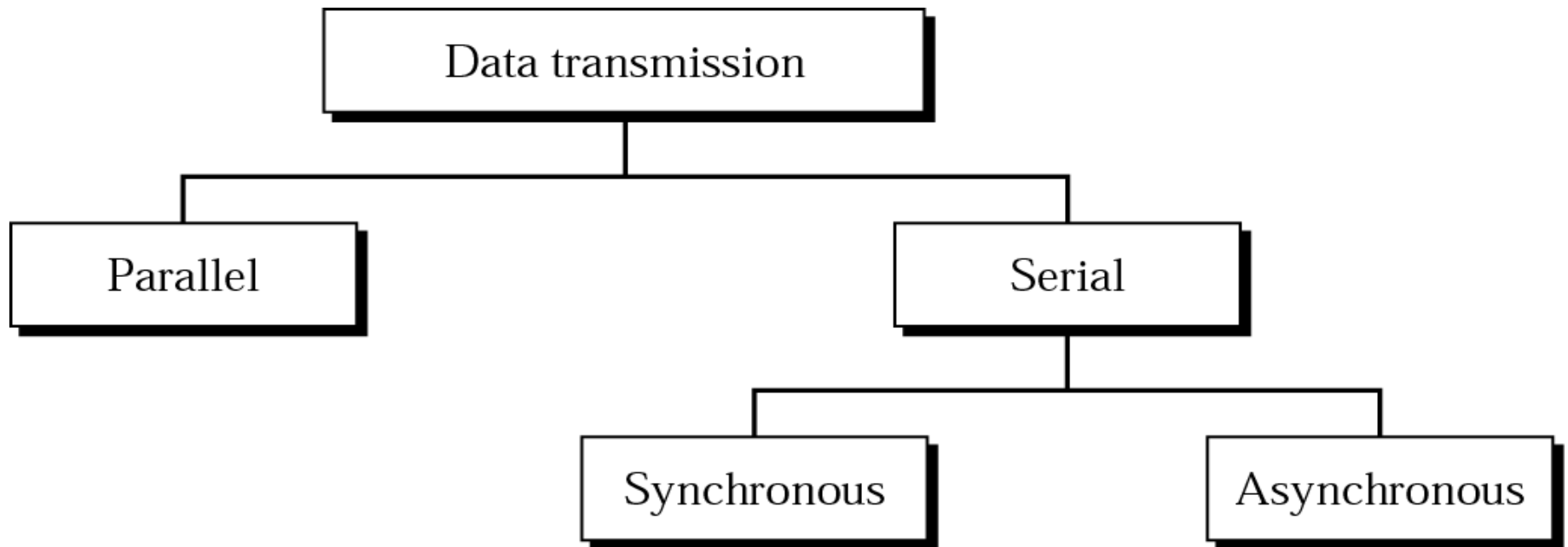


Transmission Mode

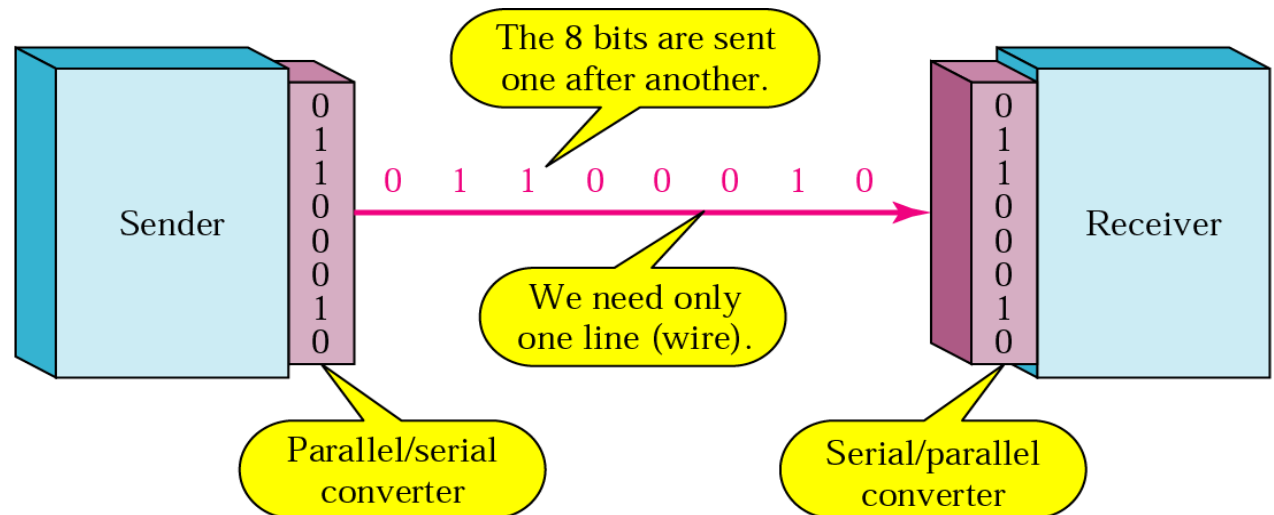
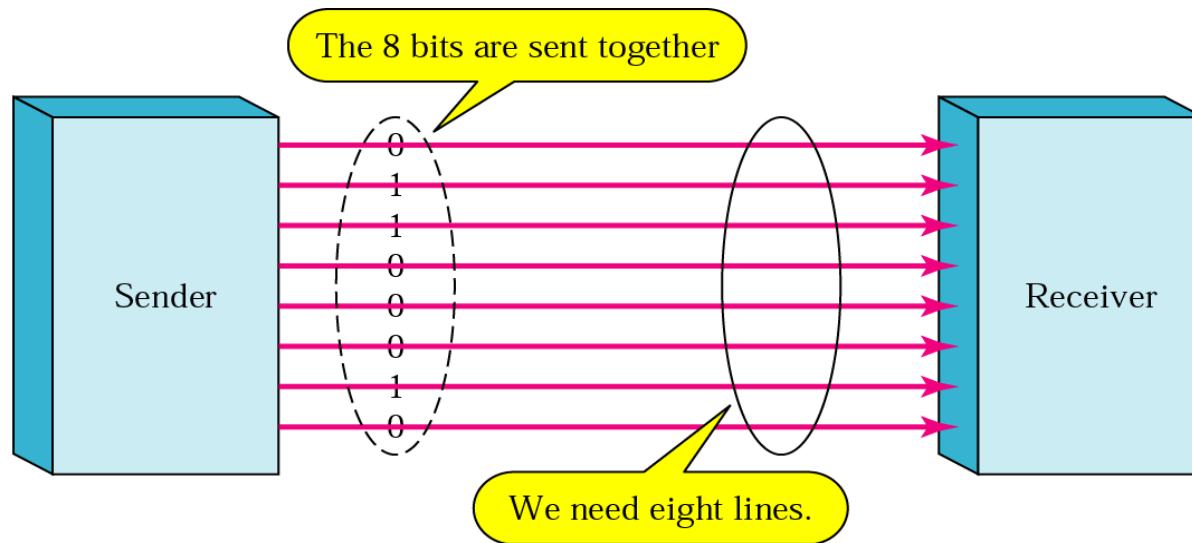
The transmission mode decides how data is transmitted between two computers.

The binary data in the form of 1s and 0s can be sent in two different modes: **Parallel** and **Serial**.

Transmission Mode



Parallel vs. Serial Transmission



Parallel vs. Serial Transmission

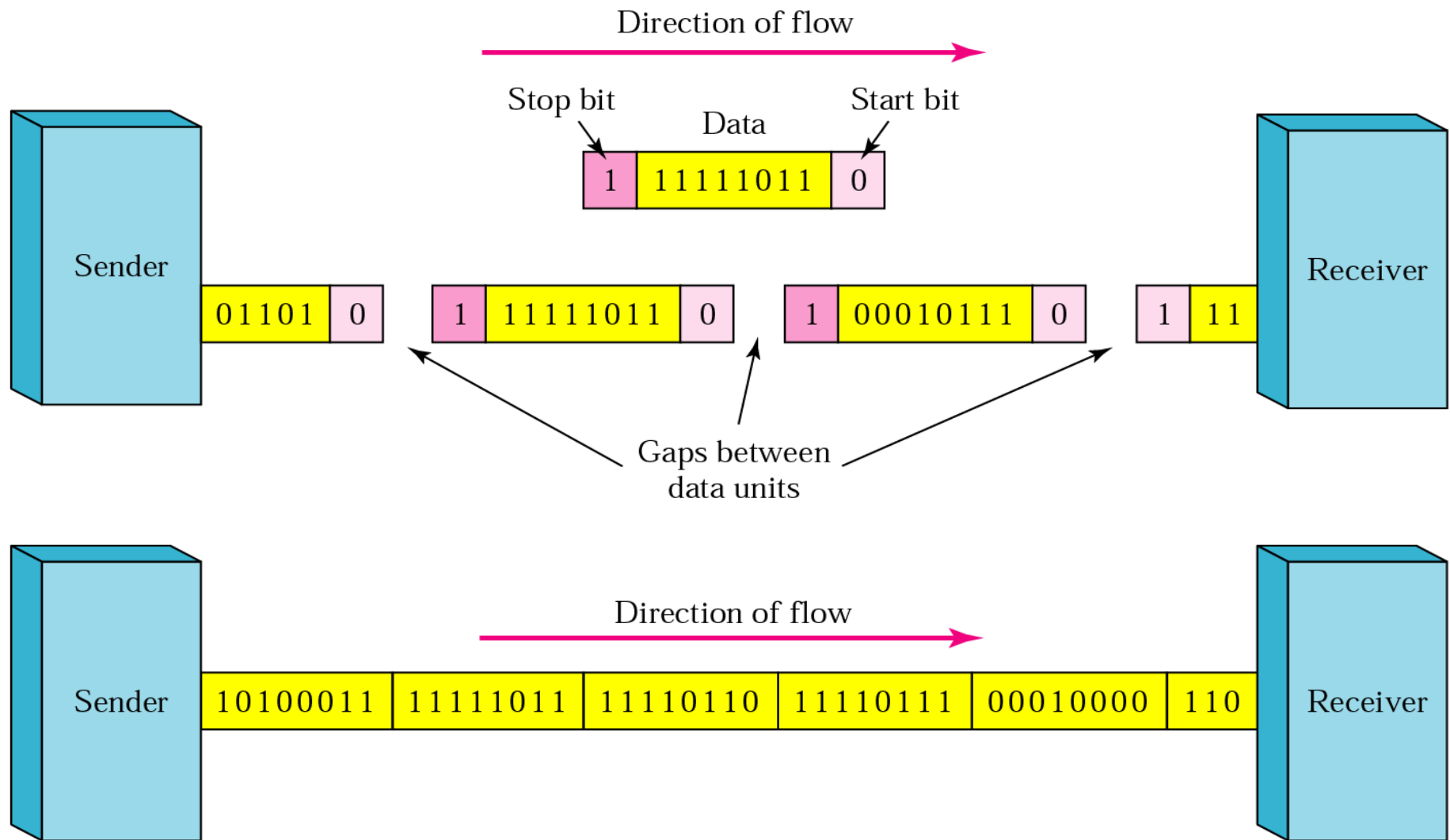
Parallel and Serial Transmission :

The binary bits are organized in-to groups of fixed length. Both sender and receiver are connected in parallel with the equal number of data lines. Both computers distinguish between high order and low order data lines. The sender sends all the bits at once on all lines. Because the data lines are equal to the number of bits in a group or data frame, a complete group of bits (data frame) is sent in one go.

Advantage of Parallel transmission is **high speed** and **disadvantage** is the cost of **wires**, as it is equal to the number of bits sent in parallel.

In **serial transmission**, bits are sent one after another in a queue manner. Serial transmission requires only one communication channel.

Asynchronous vs. Synchronous





Note:

In asynchronous transmission, we send 1 start bit (0) at the beginning and 1 or more stop bits (1s) at the end of each byte. There may be a gap between each byte.

Asynchronous here means “asynchronous at the byte level,” but the bits are still synchronized; their durations are the same.



Note:

***In synchronous transmission,
we send bits one after another without
start/stop bits or gaps.
It is the responsibility of the receiver to
group the bits.***

Questions?

Questions for exams

1. Define digital transmission, line coding.
2. Describe some characteristics of line coding: signal level vs. data level, Pulse Rate vs. Bit Rate, DC (direct-current) Component, and Self-synchronization.
3. Maths from Pulse Rate vs. Bit Rate.
4. Describe line coding schemes with examples: unipolar, polar (NRZ, RZ, Manchester, Differential Manchester), bipolar (AMI and pseudoternary).
5. Define PCM, sampling interval, sampling rate.
6. Maths from sampling rate.
7. Classification of transmission mode: Parallel and serial with example.
8. Describe Asynchronous and Synchronous transmission mode with example.

Thanks All